

TURGAY, Kazakhstan

WMO: 353580

Lat: 49.6347N

Lon: 63.4822E

Elev: 133

StdP: 99.74

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-27.3	-33.7	0.2	-29.5	-31.0	0.2	-27.2	12.0	-10.4	9.3	-15.7	2.7	20	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	36.0	20.0	34.1	19.4	32.3	18.8	20.7	33.8	19.9	32.5	19.1	31.3	2.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	11.3	27.0	14.5	10.5	26.7	13.4	9.8	26.2	60.4	33.8	57.3	32.6	54.7	31.5	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.6	7.0	-31.2	39.4	2.6	1.8	-33.1	40.7	-34.6	41.8	-36.1	42.8	-38.0	44.1		
(5)			-31.3	22.4	2.6	1.4	-33.2	23.4	-34.7	24.2	-36.2	25.0	-38.1	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-15.7	-15.0	-4.6	9.1	17.2	22.4	24.0	23.0	16.4	7.4	-2.3	-11.4		
	DBStd	15.53	7.05	6.81	7.04	5.84	4.39	4.11	3.51	4.21	4.82	4.94	6.45	7.14		
	HDD10.0	3186	796	699	454	85	3	0	0	0	6	110	369	664		
	HDD18.3	5048	1054	933	712	278	74	11	3	11	92	339	619	922		
	CDD10.0	1720	0	0	1	58	227	373	433	402	197	30	0	0		
	CDD18.3	542	0	0	0	1	40	134	178	154	33	1	0	0		
	CDH23.3	6270	0	0	0	26	462	1593	2074	1755	352	8	0	0		
	CDH26.7	2831	0	0	0	2	147	733	986	834	128	1	0	0		
(14) Wind	WSAvg	2.7	2.7	2.9	2.8	2.8	3.1	2.9	2.8	2.5	2.5	2.5	2.3	2.5		
(15) Precipitation	PrecAvg	209	13	11	13	18	15	19	18	17	14	22	21	20		
	PrecMax	312	38	30	45	58	35	110	73	60	43	78	55	47		
	PrecMin	100	1	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	49	10	8	9	17	10	24	17	17	13	18	16	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	0.8	16.3	26.2	32.7	37.6	38.4	38.1	34.2	24.2	11.7	2.1		
		MCWB	-0.8	-1.1	8.8	15.6	18.4	20.2	20.6	20.0	19.1	13.8	7.4	-0.4		
	2%	DB	-0.7	-0.7	10.3	23.1	30.2	35.2	35.8	35.5	30.2	21.1	9.0	0.7		
		MCWB	-2.0	-2.5	5.6	14.1	17.6	19.4	20.0	20.0	17.2	12.1	5.7	-0.9		
	5%	DB	-2.3	-2.5	6.3	20.7	28.1	33.0	33.6	33.5	27.2	18.3	6.7	-0.3		
		MCWB	-3.6	-4.1	3.2	12.9	16.7	18.9	19.3	19.3	16.0	10.7	3.9	-1.8		
	10%	DB	-4.8	-4.8	3.6	18.3	25.8	30.8	31.7	31.3	24.7	15.6	4.7	-1.7		
		MCWB	-6.0	-6.0	1.1	11.8	15.7	18.1	18.6	18.4	14.8	9.3	2.5	-3.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	-0.9	9.5	16.4	19.2	21.9	22.3	21.6	19.7	14.6	8.5	0.3		
		MCDB	0.3	0.8	15.6	24.8	30.1	35.5	35.2	34.9	32.7	23.7	11.2	1.4		
	2%	WB	-2.0	-2.5	6.1	14.7	18.0	20.2	20.7	20.5	17.7	12.5	5.9	-0.7		
		MCDB	-0.8	-0.8	10.1	22.2	29.6	33.0	33.6	33.6	29.4	20.1	8.5	0.6		
	5%	WB	-3.6	-4.0	3.3	13.4	17.0	19.2	19.8	19.6	16.2	11.0	4.1	-1.9		
		MCDB	-2.3	-2.5	6.2	19.9	27.0	31.8	32.1	32.3	26.5	17.7	6.3	-0.4		
	10%	WB	-5.9	-6.0	1.1	11.8	15.9	18.3	19.0	18.8	15.0	9.5	2.4	-3.2		
		MCDB	-4.8	-4.8	3.4	18.1	25.4	30.2	30.9	30.8	24.3	15.4	4.7	-1.8		
(35) Mean Daily Temperature Range	MDBR		7.4	8.1	8.6	10.9	12.3	12.5	12.3	12.3	12.1	10.6	6.6	7.1		
	5% DB	MCDBR	6.5	8.3	11.2	14.3	14.6	14.9	14.8	14.9	14.9	15.1	8.6	7.4		
		MCWBR	6.1	7.6	8.1	8.4	7.3	6.3	6.2	6.5	7.2	9.3	6.4	6.1		
	5% WB	MCDBR	6.8	8.1	11.0	13.6	13.9	14.2	13.7	14.2	14.4	14.2	8.0	6.6		
		MCWBR	6.4	7.4	8.1	8.7	7.4	6.7	6.4	6.7	7.5	9.1	6.3	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.333	0.397	0.424	0.420	0.417	0.415	0.412	0.363	0.351	0.340	0.312		
	taud		2.078	2.018	2.014	2.087	2.178	2.246	2.268	2.254	2.342	2.291	2.191	2.112		
	Ebn at Noon		698	777	789	817	839	844	839	821	830	764	659	631		
	Edh at Noon		86	117	142	147	140	132	127	123	102	89	78	73		
(44) All-Sky Solar Radiation	RadAvg		1.49	2.68	3.91	4.97	6.45	6.84	6.45	5.66	4.31	2.52	1.23	0.97		
	RadStd		0.21	0.27	0.44	0.43	0.37	0.37	0.36	0.40	0.25	0.26	0.14	0.14		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page